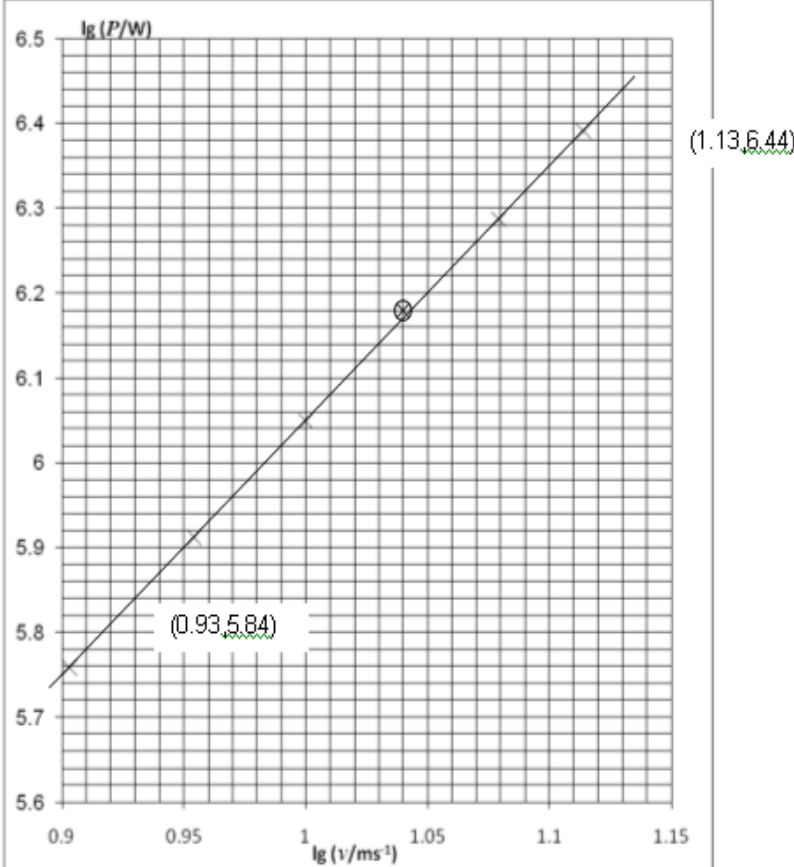


8(a)	The generator produces alternating current . The information provided in Fig. 8.3 indicates frequency of 50 Hz which is irrelevant for direct current.	[1]
(b)	Upthrust on Hywind structure, W = weight of the sea water displaced $= 1025 \times 5300 \times 9.81$ $= \mathbf{5.33 \times 10^7 \text{ N}}$	[1] ans
(c)(i)	The rotor blades exert a force on the wind to change its momentum . By Newton's 3 rd law, there is an equal and opposite force acting on the rotor blades . The force on each blade causes a moment which results in a revolution of the rotor blade about the axis. OR There is force on blades by wind. The angled blades cause/provide a component of force in plane of rotation. The force on each blade causes a moment about the axis.	[1] [1] N3L [1]

8(c)(ii)	The kinetic energy of the wind is used to do work in turning the coil in the generator. The gear box coupled to the generator cause the coils in the generator to rotate and cut the magnetic flux inside the generator. The cutting of the magnetic flux by the rotating coil in the generator produces electrical energy through the effect of electromagnetic induction.	[1] [1]
(d)	Using points: $(10.5, 1300000): k = \frac{1300000}{10.5^3} = 1123 = 1120$ $(11.5, 1700000): k = \frac{1700000}{11.5^3} = 1118 = 1120$ $(14.5, 3400000): k = \frac{3400000}{14.5^3} = 1115 = 1120$ k is approximately 1120 for all the 2 points tested. Hence we can conclude that k is a constant.	[1] working Any 2 [1] reason
(e)(i)1.	When $v = 11 \text{ m s}^{-1}$, $P = 1500000 \text{ W}$ $\lg (11/\text{m s}^{-1}) = \mathbf{1.04}$, $\lg (1500000/\text{W}) = \mathbf{6.18}$	[1] correct plot
2.		[1] best fit line drawn
(ii)	Using points from the graph: (0.93,5.84), (1.13,6.44) $\text{Gradient} = \frac{6.44 - 5.84}{1.13 - 0.93}$ $= \mathbf{3.0}$	[1] sub [1] ans

8(e)(iii)	Given $P = kv^3$ $\lg P = 3 \lg v + \lg k$ where gradient of the line is 3. Since the experiment data shows a straight line graph with gradient equals 3.0, the relationship $P = kv^3$ is valid.	[1] exp [1] comment with reason
MC	ecf for comment	
(f)(i)	Kinetic energy of air incident per second, $P = \frac{1}{2} \rho A v^3$ $= \frac{1}{2} (1.2) \pi \left(\frac{82.4}{2} \right)^2 (11.0)^3$ $= 4.26 \times 10^6 \text{ W} = \mathbf{4.26 \text{ MW}}$	[1] sub [1] ans
(ii)	Overall efficiency = $\frac{\text{Power output}}{\text{Power input}} = \frac{1500000}{4.26 \times 10^6} = \mathbf{0.352}$	[1] ans
(g)	Wind speed which is excessively high leading to over current produced which will damage the generator. Rotor blades struck by lightning since they are the tallest point of the wind turbine and they have relatively pointed ends. Corrosion of the structure due to the salty environment. Power loss due to long distance transmission. Difficulty in maintenance as it is out in the open sea.	[2] any 2